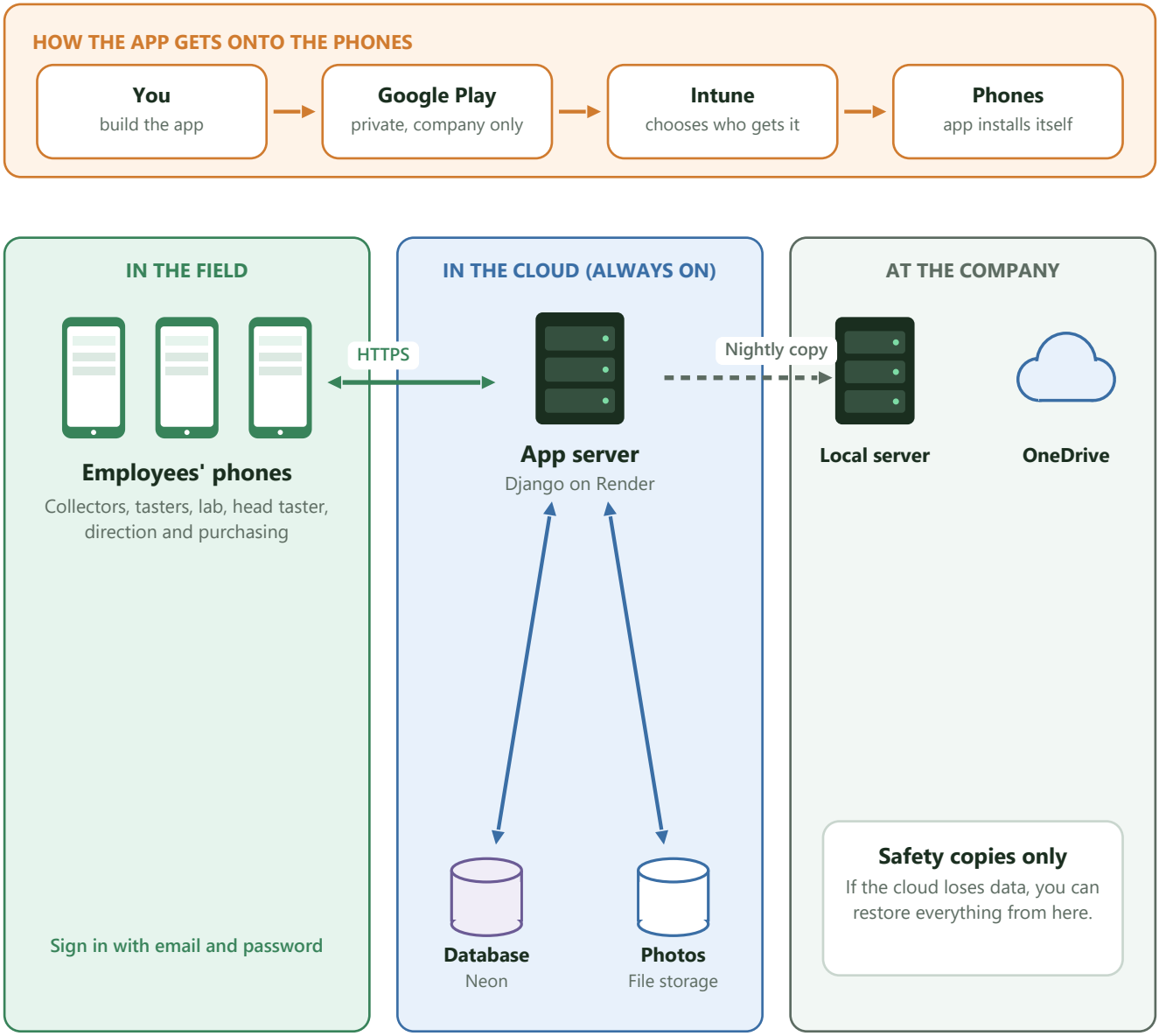
AL JAZEERA STCA

Company deployment plan

How to get the app onto every employee's phone privately, without a public store listing, using what the company already has.



The plan in one sentence: the server runs in the cloud, employees get the app privately through Intune and Google Play, and the company keeps a safety copy of the data every night.

Written from the project on 31 October 2026. The pictures in this guide are diagrams, not screenshots. Menu names in Microsoft and Google websites can change.

START HERE

What each thing is

Each service has one job in the plan.

Domain

The company's name on the internet.

Gives your server its address: api.oliveoiltunisia.com

Microsoft 365

The company's email and work accounts.

Holds the employee accounts Intune needs.

OneDrive

Online file storage, like Google Drive.

Keeps backup copies. It cannot run your server.

Local server

A computer that stays on at the company.

Receives a safety copy of your data every night.

Intune

Microsoft's remote control for phones.

Decides which phones receive your app.

Google Play (private)

A private shop only your company sees.

Stores the app. The public cannot find it.

Render

A company that rents always-on servers.

Runs your Django server, day and night.

Neon

A company that hosts databases.

Stores samples, users and evaluations.

Only two decisions

QUESTION 1

How does the app reach the phones?

Answer

Intune + private Google Play

QUESTION 2

Where does the server run?

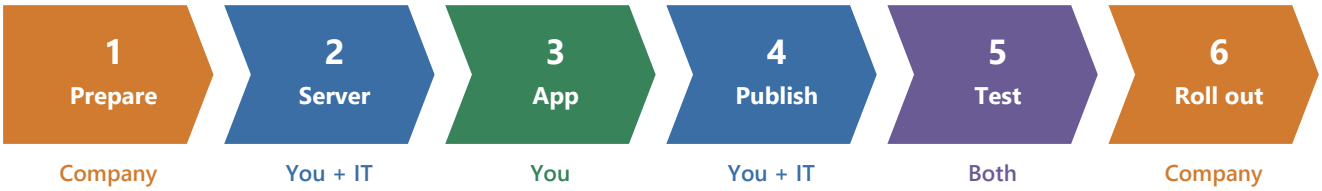
Answer

Cloud, with backups at the company

THE ROADMAP

Six phases, in order

Each phase depends on the one before it. Do not invite employees until Phase 5 works.



The colour shows who does most of the work.

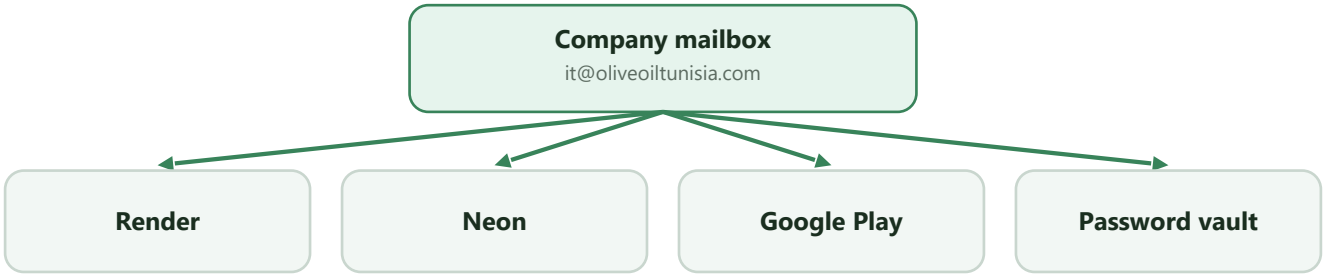
Phase	What happens	Finished when
1. Prepare	The company confirms Intune and creates its accounts.	All accounts exist in the company's name.
2. Server	The cloud server becomes the real production server.	The app address opens over HTTPS and the backup restore works.
3. App	The app gets its final identity and signature.	A signed build passes all tests.
4. Publish	The app goes into the private store and Intune.	Intune can assign the app to a group.
5. Test	2 or 3 real people use it on real phones.	One sample completes the whole journey and an update arrives.
6. Roll out	Employees join group by group.	Everyone is signed in and knows who to call.

Two things still need confirming: whether the Microsoft plan includes Intune, and whether the company accepts its data in the cloud. Phase 1 answers both. If the company prefers the local server, only Phase 2 changes.

PHASE 1 · THE COMPANY

Prepare the company

Nothing technical yet. This phase makes sure the company, not one person, owns everything.



1. **Find the Microsoft 365 administrator**
   Ask who manages the company's Microsoft account. That person signs in at **admin.microsoft.com** and opens **Billing > Your products**. Business Premium, E3 and E5 include Intune. With Business Basic or Standard, add Intune or upgrade, starting with licenses for the pilot only.

2. **Name one person responsible**
   Choose an IT employee or an outside provider who will add phones and manage access. You should not be the only person who can do this.

3. **Create a shared company mailbox**
   For example **it@oliveoiltunisia.com**. Use it for every service account, so access stays with the company when someone leaves. Confirm the exact domain name with the administrator.

4. **Create the service accounts**
   Create or transfer Render and Neon to that mailbox, and register a Google Play developer account as an **organization**. Google asks for company documents and may ask for a D-U-N-S number, so start early. The one-time fee is about $25.

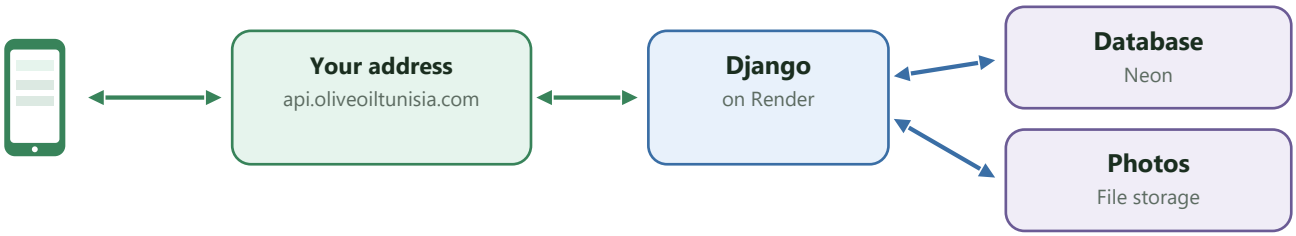
5. **Get access to the domain settings**
   Find who can edit the settings for the domain. You will need one new record in Phase 2.

> **Ask the company:** "Does our Microsoft 365 plan include Intune? Who can edit the domain settings? Do you accept having the app's data in the cloud, with a nightly copy here?"

PHASE 2 · YOU + IT

Set up the server

Turn the demo server into a real one that stays on and keeps data safe.



6 **Move to paid plans**
Upgrade Render and Neon so the server does not fall asleep and the database has enough space.

7 **Set the production values**
In Render's environment settings, set **DEBUG=False**, a new **SECRET_KEY**, the real **ALLOWED_HOSTS** and **DATABASE_URL**. Never put these in the code.

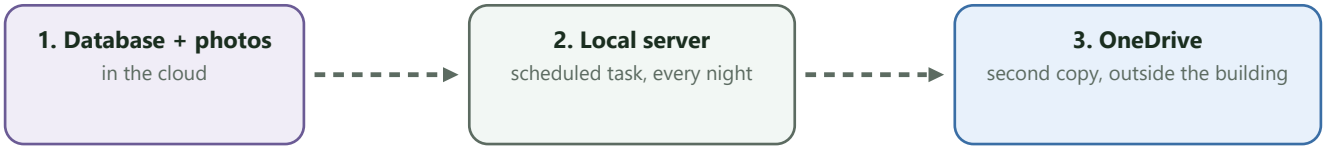
8 **Give the photos safe storage**
Today the photos sit on the server's own disk, which Render can erase on each deployment. Use external file storage. It needs a small code change, and it is easier to back up than a Render disk.

9 **Connect the address**
Add **api.oliveoiltunisia.com** in Render, then add the record Render gives you to the domain settings. Render then provides HTTPS. Open the address in a browser to confirm.

10 **Set up the password reset email**
Connect a company mailbox so "Forgot password" can send codes, then test it with a real account.

11 **Prepare the real data**
Run **python manage.py migrate**, delete every test account such as direction@stca.tn, and create one account per person with a real email.

The nightly backup

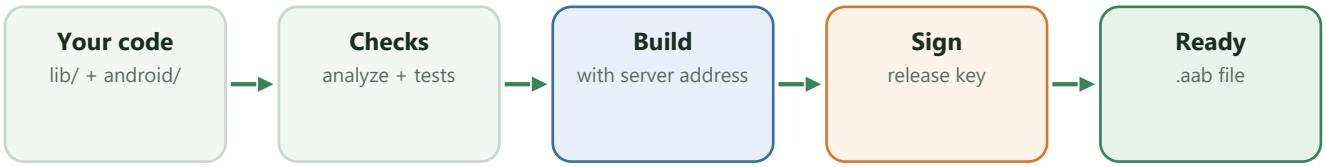


12 **Automate the backup, then test a restore**
Schedule a nightly task on the local server that saves the database with **pg_dump** and copies the photos. Keep a second copy in OneDrive. Then restore it into a test database: a backup you have never restored is not proven.

PHASE 3 · YOU

Prepare the app

Give the app its final identity. Mistakes here are the hardest ones to undo later.



13 Change the app ID

In **android/app/build.gradle.kts**, line 24, replace **com.example.project3** with **com.oliveoiltunisia.stca**. Confirm it first, because changing it later creates a new app.

14 Create the release signing key

This key proves that updates really come from you. Keep it out of git, then replace the debug signing on line 37 of the same file.

```
keytool -genkey -v -keystore stca-release.jks -keyalg RSA -keysize 2048 -validity 10000 -alias stca
```

15 Turn off unencrypted connections

In **android/app/src/main/AndroidManifest.xml**, line 8, set **usesCleartextTraffic** to false. Keep it true only in a debug manifest, so local development still works.

16 Add a minimum version check

When the app opens, it asks the server for the lowest allowed version. If the phone is too old, it shows a clear "please update" screen. This stops old versions from working against a newer server.

17 Run all the checks

All must pass before you build.

```
flutter analyze
flutter test
cd backend_new
python manage.py test
python manage.py makemigrations --check --dry-run
```

18 Build the app

This places the real server address inside the app.

```
flutter build appbundle --release --dart-define=API_BASE_URL=https://api.oliveoiltunisia.com
```

Protect the signing key: keep it in the company's password vault, with a second copy stored separately. If you lose it, the installed app can no longer be updated normally.

PHASE 4 · YOU + IT

Publish privately

Put the app in the company's private store, then decide who receives it.



19 **Connect Intune to Managed Google Play**
The administrator signs in at **intune.microsoft.com**, opens the Android enrollment settings and connects the company's Managed Google Play account.

20 **Find the Organization ID**
It is shown in Managed Google Play. Google uses it to know which company may see the app.

21 **Create the app as private**
In Google Play Console, create the app, turn on the restriction for your organization, add the Organization ID and upload the build. Once restricted, the app is available only to that organization.

22 **Bring the app into Intune**
In Intune, sync or approve the app from Managed Google Play so it appears in the app list.

23 **Create the groups**
Create Collectors, Tasters, Head taster, Lab, Direction and Purchasing. Assign the app to a group, not person by person.

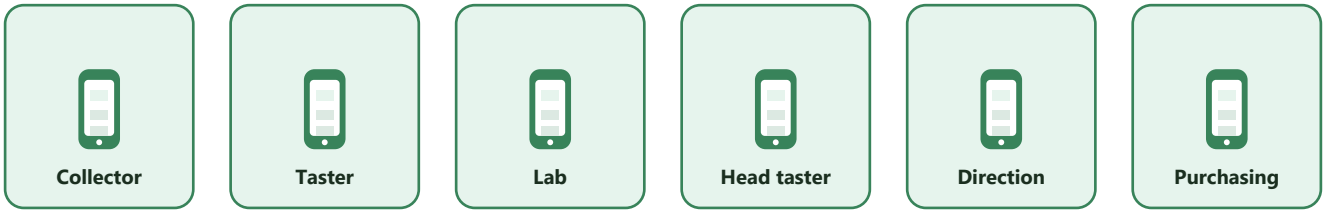
Who owns the phone	How to add it	What the company controls
The company	IT scans a QR code during setup.	The whole phone.
The employee	They sign in with a company Microsoft account, which creates a work profile.	Only the work section, not personal photos or messages.

If a page asks for an .apk: Google's guidance differs by upload method. Build it with the same command, using **apk** in place of **appbundle**.

PHASE 5 · COMPANY + YOU

Test with real people

Find problems with 2 or 3 phones, before they affect the whole company.



Every role should handle the same real sample in its normal way of working.

24 **Add the pilot phones to Intune**
Choose people from different roles, including at least one collector who works outside the office.

25 **Assign the app**
Confirm that it installs without help, and that the app reaches the server on mobile data as well as Wi-Fi.

26 **Run one real sample from start to finish**
Go from collection through tasting, lab analysis and the purchase decision. Check that each person sees only what they should, with no status or work from one user reaching another.

27 **Fix problems, then release one small update**
Raise the version, upload it and confirm it reaches the pilot phones. This proves your update process before you depend on it.

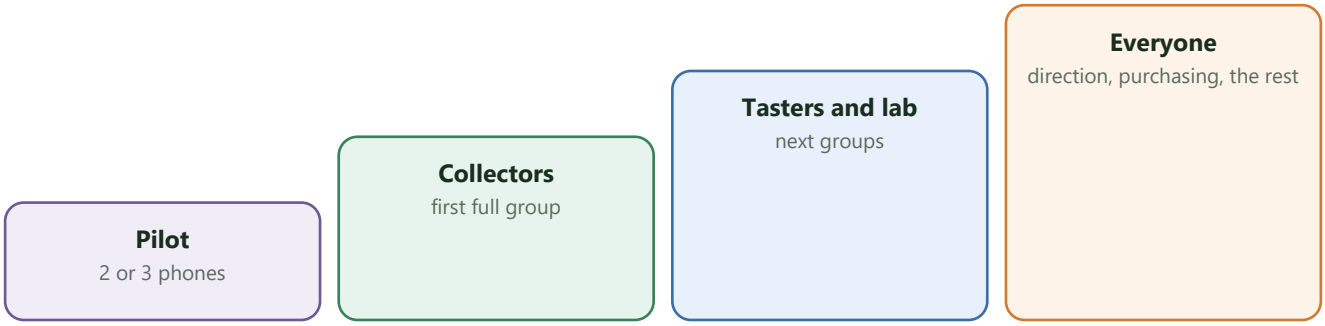
Ready to continue when:

- [ ] The app installs on every pilot phone
- [ ] Signing in and "Forgot password" both work
- [ ] A real sample completes the whole journey, with its photos
- [ ] An update reached the phones
- [ ] Last night's backup exists and has been restored once

PHASE 6 · THE COMPANY

Roll out to everyone

Add people in waves, so you can help each group properly.

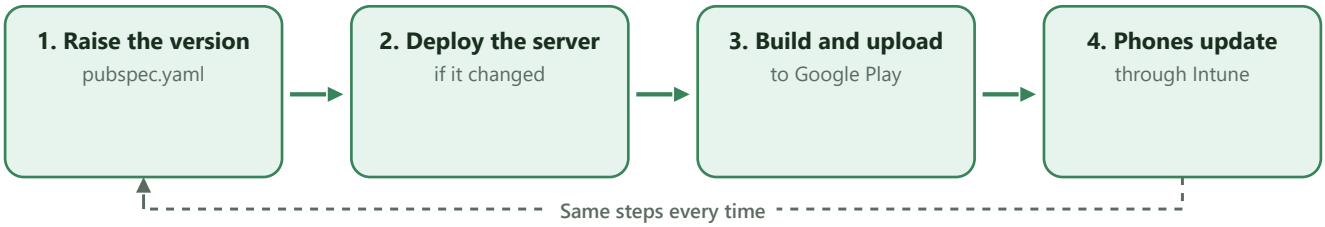


28 **Add one group at a time**
Start with the group that uses the app most. Wait a few days, fix what appears, then add the next.

29 **Give each person what they need**
Their account, a one-page guide and a short demonstration.

30 **Name who to contact**
Tell everyone who to call when something fails, and how to reach them while travelling.

Releasing updates after launch



31 **Raise the version**
In **pubspec.yaml**, the number after the + must always increase, for example from **1.0.0+1** to **1.0.1+2**.

32 **Deploy the server first**
Any server change must still work with the app version on people's phones.

33 **Build and upload**
Upload the new version to Google Play. Intune then delivers it to the assigned phones.

SPECIAL CASES

Gmail accounts and iPhones

Signing in to your app

Gmail works

Any real email the person reads. Password reset codes go there.

Installing through Intune

Gmail alone does not

The phone must be managed by Intune. Personal phones need a company Microsoft account.

Person	What to do
Collector with a company phone	IT adds the phone to Intune. They can still sign in to the app with Gmail.
Collector with a personal phone	Give them a company Microsoft account with an Intune license, used only to set up the phone.
Someone outside the company's systems	Send the app through Firebase App Distribution, which works with a Gmail address.
Anyone with an iPhone	This guide covers Android. iPhones need an Apple Developer account and a Mac to build the app.

What it may cost

Estimates only. Check the current prices before buying.

Item	Rough cost	Note
Google Play developer account	About $25, one time	Required for the private app.
Microsoft Intune	Around $8 per user each month	May already be included in the company's plan.
Render and Neon	About $5 to $20 each month to start	Grows with usage, so confirm after the pilot.
Domain, OneDrive, local server	Already owned	No new purchase expected for this plan.

KEEP THIS PAGE

Master checklist

Tick each item as it is finished.

Phase 1: Prepare

- [ ] Intune confirmed or added
- [ ] Responsible person named
- [ ] Company mailbox created
- [ ] Render, Neon and Google Play accounts owned by the company
- [ ] Domain access obtained

Phase 2: Server

- [ ] Paid plans active
- [ ] Production values set
- [ ] Photos in safe storage
- [ ] HTTPS address working
- [ ] Password reset email tested
- [ ] Test accounts deleted, real accounts created
- [ ] Nightly backup running and restore tested

Phase 3: App

- [ ] App ID changed
- [ ] Release key created and stored twice
- [ ] Unencrypted connections turned off
- [ ] Minimum version check added
- [ ] All tests pass
- [ ] Signed build created

Phase 4: Publish

- [ ] Intune connected to Managed Google Play
- [ ] App published privately
- [ ] App visible in Intune
- [ ] Groups created

Phases 5 and 6: Test and roll out

- [ ] Pilot phones working on Wi-Fi and mobile data
- [ ] One real sample completed
- [ ] One update delivered
- [ ] All groups added
- [ ] Support contact shared with everyone